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1. Consider the nonlinear equation $y = f(x) = x^2 + 2$, where the input x is a Gaussian random variable having mean of 1 and variance of 1. We desire to calculate the statistics of the output variable y using the approach used by EKF in this quiz (and then use a different approach in the next quiz).

1 / 1 point

If we approximate $\mathbb{E}[f(x)]$ with $f(\mathbb{E}[x])$, what value do we find for $\mathbb{E}[y]$?

3

✔ Correct

Yes. That is the right answer. Note, however, that the true expected value of y is actually 4. The EKF approach fails!

2. For the same problem described in question 1, if we desire to calculate the covariance of y using the EKF approach, we must first linearize $f(x)$. What is the value of $\hat{A} = \partial f(x) / \partial x$, evaluated at $x = \bar{x}$?

1 / 1 point

2

✔ Correct

Yes. That is the correct result.

3. For the same problem described in question 1, we would calculate $\Sigma_{\tilde{y}} = \hat{A}\Sigma_{\tilde{x}}\hat{A}^T$ if we were using the EKF approach. What is the value you compute for $\Sigma_{\tilde{y}}$ using this method?

1 / 1 point

4

✔ Correct

Yes. That is the correct answer to this question. The EKF approximates $\Sigma_{\tilde{y}}$ as 4, but the exact value for $\Sigma_{\tilde{y}}$ if we do all the difficult integral math is actually 6. The EKF approach fails!